

Mental Health and Strength and Conditioning

Mental health is emerging as an important topic for strength and conditioning professionals. Research published related to mental health and athletes has accelerated from roughly 74 publications in 2003 to over 1,100 in 2023. This growth demonstrates increased interest in the topic, but research outcomes support the need to focus on building competence in strength and conditioning because mental health affects performance (24, 62). In a study, student-athletes were at risk for depression (22.3%), anxiety (12.5%), and low self-esteem (8%) (62). While most research has reported higher rates of positive mental health for athletes compared to nonathletes, training for and playing sports are stressful (31, 47). Some of the stressors include burnout from long hours of training; social identity pressure to perform well; and meeting high expectations from coaches, teammates, and self while attempting to compete with other athletes. These stressors can affect mental health and hygiene that can appear at subclinical levels in student-athletes.

What is mental health? Defining mental health encompasses emotional, psychological, and social well-being (5, 47, 62). Also, mental health is not just the absence of a mental illness. Challenges to mental health are often detected when sport, training, and life challenges disrupt normal self-care behavior. Coaches working with athletes will have varied backgrounds related to psychological distress identification and mental health management. Some practical parameters for coaching education related to mental health issues are needed.

Strength and conditioning professionals can observe signs and symptoms of changing mental health in athletes. While striving to enhance performance, a holistic athlete monitoring program can address self-care, recovery, and optimization of psychological, behavioral, and emotional factors (5).

Strength and conditioning professionals can view psychological distress in many ways. One method is for coaches to document issues in mental health through identifying signs and symptoms ([table 9.1](#)). Signs are observable outcomes that can be readily assessed by the coach. For instance, a sign of depression is the lessening of effort observed by the coach. Symptoms are usually self-reported feelings or sensations. The symptoms would be the athlete reporting that he or she feels tired or has less power.

TABLE 9.1 Signs and Symptoms of Anxiety, Depression, and Stress

Signs		Symptoms	Distinctive factors
Anxiety	• Rapid eye movement • Rapid speech • Sweating • Racing heart rate • Shaking or tapping with fingers or toes • Sleep problems	• Feeling restless, wound up, on edge, or irritable • Easily fatigued • Difficulty concentrating • Gastrointestinal somatic symptoms • Difficulty controlling feelings of worry • Feelings of impending doom or being out of control • Feeling self-conscious or avoiding situations	• Circular thinking, rumination, worry
Depression	• Depressed mood • Sleep disruption • Decreased movement or coordination • Agitation • Social isolation	• Change in appetite • Feelings of guilt • Gastrointestinal somatic symptoms • Hypochondriasis • Rumination about the past	• Low energy or mood, hopelessness, and helplessness
Stress	• Racing heart rate • Agitation • Sleep disturbance	• Concerns about meeting high standards • Concerns about social obligations	• Urgency or heightened arousal

Adapted from Beisecker et al. (2024); Gustavson et al. (2018); Knowles et al. (2021); Rogers et al. (2024); Weber et al. (2023).

Signs	Symptoms	Distinctive factors
• Inability to recover • Appetite changes • Impulsive decision making	• Concerns about wasting time • Concerns about weight • Trouble making decisions • Trouble relaxing • Being lonely	

Adapted from Beisecker et al. (2024); Gustavson et al. (2018); Knowles et al. (2021); Rogers et al. (2024); Weber et al. (2023).

A clear understanding of similarities and differences between mental states can be useful. It is important to note that several nuances of anxiety exist and can include general anxiety disorder, social anxiety, panic disorders, and phobias. Anxiety often presents as a circular pattern of thoughts and feelings of heightened importance known as *ruminations* and *worry*. Some predisposing factors for anxiety include large social and schedule adjustments, a biological family history of anxiety, negative life events, early trauma, and shyness in childhood (6, 62). Lists of signs and symptoms of anxiety, depression, and stress in athletes are in [table 9.1](#). These manifest in varied somatic, cognitive, and behavioral symptoms. Often maladaptive coping and perceptions of impending outcomes or doom fill the individual with urgency, overwhelming feelings of dread, and focus on negative consequences. By the end of this section, strength and conditioning professionals will be able to identify signs and symptoms of anxiety, depression, and stress; recognize the appropriate means of support; and develop an appreciation of the impact of mental health on performance and injury risk.

Depression and anxiety can share signs and symptoms. These include agitation, somatic complaints (e.g., stomach aches or sweaty palms), restlessness, social withdrawal, difficulty thinking, and sleep disturbance. These represent cognitive processing biases that can lead to rumination and worry (26). What sets depression apart is

anhedonia (lack of emotion), feelings of hopelessness or helplessness, and low energy (26). Studies have identified that depression can begin with poor performance, injury, concussion, social comparisons, high expectations, and changes in social or playing status (26, 31). Also affecting depression are changes in feedback from coaches, behavioral deactivation, negative self-perceptions, evaluations, and feelings of helplessness or hopelessness. Stress is an imbalance between the demands of a task or situation and the ability to cope with the challenge, as stated earlier in this chapter. Chronic stress often interferes with sleep, mood, and self-care habits. Because mental stress is unique to each individual, it has many signs and symptoms. The most recognizable signs of stress include urgency, agitation, and hurried speech.

Now that these states have been identified, how do coaches help athletes get the appropriate support? One helpful way is a three-step process that follows the content of Weinberg and Gould (64) while adding the assessment of duration and disruption from more traditional psychological evaluation. First, identify the signs and symptoms of distress related to affect, behavior, and cognitions. Second, determine the duration of distress (lasting greater than 2-3 weeks) and potential disruption (e.g., sleep patterns, nutritional habits, social relationships). Specifically, is the distress temporary or persistent, and how will it affect behaviors outside of sports? Third, determine how to support the athlete. A question could follow: Is it important to refer the athlete for more support, allow the athlete time for self-reflection, or collaborate with the athlete to work on centering or environmental management to change disruptive performance-related responses?

The first step in identifying psychological signs of distress is classifying the affect, behavior, and cognitions (ABCs) observed in athletes (11, 64, 65). *Affect* refers to the mood and emotions seen in and reported by the athlete. For instance, an athlete might look anxious, have a stressed face, report feeling hopeless, or have a persistent negative expression. *Behavior* refers to movements like

nervous pacing, fidgeting, slow reactions, and rapid eye tracking. *Cognitions* are thoughts reported, negative worry about performances or outcomes, overgeneralizing, catastrophizing, and doubts about abilities or past failures. A brief review of cognitions that can disrupt performances are all-or-nothing thinking, jumping to conclusions, magnification, minimization, emotional reasoning, and confirmation bias (5, 22).

Once these signs are observed, step 2 is to determine if the issue is temporary or persistent. To determine what level of help is required for an athlete, coaches need to learn how the signs and symptoms of distress affect performance and when and how to refer to a professional who is trained to handle psychological distress.

This third step, referral to a qualified support professional, requires the coach to work in an individual setting with the athlete to develop a plan to refer to a support professional, who should be consulted or contacted ahead of time. This remains a key skill to assist athletes with self-care for mental health and as an important step for coaches. A performance coach or consultant is trained in implementing psychological skills that are attention based and do not enter into interpersonal or clinical issues. For example, a performance coach or consultant (usually certified through the Association of Applied Sport Psychology as a certified mental performance consultant [CMPC]) employs goal setting, imagery, self-talk, attentional focus, or other psychological skills to direct effort and attention. A counselor is licensed by the state and is usually trained to help with interpersonal, cultural, and mental health issues. For example, if an athlete needs a counselor, interpersonal and self-care can become a focus of cognitive behavioral treatment. Also licensed by the state, a psychologist is trained to administer psychological testing batteries, focusing on interpersonal and self-care issues to work with chronic mental health issues.

The *Mental Health Case Studies* sidebar discusses different issues coaches encounter and highlights the referral process. When issues relate to performance of low motivation and effort, coaching cues

are often given. However, when the athlete demonstrates persistent (two to three weeks) negative affect, cognition, or behaviors, referral for more support to a counselor or psychologist may be needed. Also, if the behavior becomes disruptive to other teammates or training partners, the coach is tasked with assessing the situation, making a judgment related to management of the athlete's behavior, and deciding how best to address or support the athlete toward positive behaviors.

In the future, challenges to mental health in athletics could also come from social media; name, image, and likeness (NIL); and temporary commitments or lack of identification with a program facilitated by the college transfer portal. Thus, a higher level of social comparison and self-esteem challenges will likely be stressors for athletes. Imperative to the preparation for future strength and conditioning professionals will be skill building and resources related to these emerging challenges.

When subclinical anxiety, depression, or stress are seen by the coach, a plan can be made to change behavior, team environment, and emotional management. Subclinical issues include low-level and short-term mental or emotional distress that are not serious enough to affect athletes' personal, academic, and social roles. These often do not last longer than a week and do not disrupt sleep, eating, and social functioning. However, when persistent issues or disruptions become more chronic and severe, a professional referral to a mental health specialist is clearly warranted. When more chronic issues arise, the coach should seek support of the sports medicine staff (e.g., athletic trainers, nutritionists, team physicians) as well as the mental health community in a respectful way (maintaining confidentiality, dignity, and communication with the athlete), so prioritizing mental well-being is the first goal. By developing a reference list of professionals in the school, surrounding community, or athletic department, the strength and conditioning professional can be better prepared for mental health issues should they arise.

Social norms and the environment should be considered when helping an athlete seek mental health support. Athletes in team sports may wish to have privacy while getting support. Because social stigmas and mental health treatment can sometimes clash, the strength and conditioning professional should attempt to be aware of individual preferences and readiness for support. Not all mental health issues are covered in this chapter (e.g., substance abuse, eating disorders, bullying or hazing, body dysmorphia, gender identity, social phobias, and trauma) so further education is recommended for strength and conditioning professionals as they work with diverse populations of athletes.

Mental Health Case Studies

The strength and conditioning professional can use the list of signs and symptoms of anxiety, depression, and stress found in [table 9.1 \(page 182\)](#) to frame how each athlete is feeling (step 1), then determine how long and how critical those signs and symptoms are to the athlete (step 2), and decide a course of action to refer the athlete to the appropriate support professional (step 3).

Case Study 1

Tanner has been swimming all summer and feels tired and demotivated. He is a high school junior, ready for his senior year, and feels like his friends are too busy to spend time with him. This has lasted for one week. He reports that no matter how hard he tries, he cannot get better. During workouts he reports not wanting to swim anymore. His workout effort is decreasing, but he still completes all sets. He paces in the water between sets and looks uncomfortable. Socially, he has not been talking to his friends during workouts but reports a good social life outside of the strength and conditioning programming.

Step 1: Signs and Symptoms

Pacing, reduced effort, lack of social interactions during the workouts, and decreased effort and motivation

Step 2: Duration and Disruption

One week in duration, not affecting things outside of sport

Step 3: Referral Source

Performance coach or consultant

Case Study 2

Jenny has been training to get better at volleyball but on her college team has good competition. She nervously fidgets with her hands before drills and often reacts without concerns for form or accuracy. Also, she will storm out of the gym when her emotions get the best of her. This has been occurring for three months. She is a dedicated athlete in the strength program. Often she feels less talented compared to her peers and pushes herself to prove everyone wrong about her nonscholarship status. At night, she often worries about her choices, her ability, and her friendships. This has caused her to lose sleep, averaging about three to four hours a night. She also starts missing workouts due to oversleeping and performs poorly in coordination drills. When approached about her effort, she gets defensive and leaves training.

Step 1: Signs and Symptoms

Fidgets and storms out of the gym, missing workouts, oversleeping

Step 2: Duration and Disruption

Three months; it seems to be disrupting other aspects of her life such as social relationships or sleep

Step 3: Referral Source

Counselor

Case Study 3

Meisha is a 15-year-old softball player who is very outgoing. She is happy but can be too intense for her teammates. They report her mood as very aggressive and confrontational. This has been an issue for the last month. Wanting attention, Meisha often wants to talk about leading drills, wants to be the first in line, and talks loudly. Her relationships, sleep, and self-care seem good, but lately friends have started to call her less. She wants to learn more about goal setting and self-talk so she can get stronger in the off-season. Coaches report several girls on the team have reported that her mood is creating an issue on the team such that teammates are starting to feel uncomfortable. Meisha reports feeling socially isolated and sad about this development.

Step 1: Signs and Symptoms

Loud talking, aggressive, feeling uncomfortable with being socially disengaged

Step 2: Duration and Disruption

One month, socially disruptive, not affecting behavior outside of sports

Step 3: Referral Source

Performance coach or consultant

Case Study 4

Alex is a high school senior who cannot seem to find his position on the soccer team. His mood keeps getting worse. This has been happening for the last six

weeks. His friends are worried about his continued drop in mood, and he has started missing school. When asked about his motivation and plans for soccer, he says he has no idea and he does not really care anymore. It is clear he is starting to get worse with his self-care, and he stops shaving. His friends have noticed him vaping after school, and he is starting to fall behind during running drills. Recently, his teammates have tried to help but are starting to get worried.

Step 1: Signs and Symptoms

Missing practices and school, mood, and decreased motivation

Step 2: Duration and Disruption

Six weeks, disrupting social relationships and self-care

Step 3: Referral Source

Counselor or psychologist

Psychological Impact of Injury in Sport

To enhance athletic performance and decrease the risk and severity of future injury, strength and conditioning professionals need to build some basic skills. Injuries are challenging setbacks many athletes face during their careers (39). Whether athletes work through a minor setback (e.g., a muscle strain or adjusting to a new season) or a career-ending injury, the impact on an athlete's psychological health and performance is significantly affected. Among the cognitive responses, athletes may experience recurrent intrusive thoughts, decreased self-esteem, and lowered **athletic identity** (identification and self-worth from sport participation) (14). Additionally, an athlete's emotional response will vary throughout the injury rehabilitation process. They will feel a spectrum of negative emotions and behaviors such as anger, fear, confusion, depression, helplessness, unreasonable fear of reinjury, denial of injury severity, and extreme guilt (1, 14). As the strength and conditioning professional spends more time with the injured athlete throughout the rehabilitative process, attempts to help develop a culture of athlete safety and support throughout the return-to-play process are designed to lessen fear and enhance motivation (59). With the emerging issues of concussion protocols and increased awareness of lingering effects from more severe injuries, strength and conditioning professionals should work to help prevent further distress to the athletes they serve.

To better support the athlete during this transition, it is essential to understand how psychological factors may affect an athlete during the pre- and post-injury timeline.

Psychological Factors That Can Predispose an Athlete to Injury

Research has suggested that a variety of psychological factors can help explain the predisposition of athletes suffering an injury (68).

The stress and injury model is a theoretical framework that attempts to explain the effects of pre-injury psychological factors on athletic injury (3). Revised by Williams and Andersen (68), the model suggests that three unique factors can influence an athlete's stress response, separately or together, contributing to an athlete's risk and frequency of injury ([figure 9.4](#)). It includes athletes'

- personality factors such as state or trait anxiety levels, mood states, and perceived locus of control of situations and events;
- history of stressors such as life events, academic stressors, daily hassles, and previous injury history; and
- coping resources such as social support, stress management techniques, mental skills, and nutritional and lifestyle habits.

When a potentially stressful athletic situation occurs during training, practice, or competition, these factors may affect how athletes respond to the situation, either positively or negatively. For example, an athlete with a history of high stress, a perfectionistic personality, and anxiety is likely to negatively evaluate an injury. In this case, the situation is more stressful and the athlete will likely exhibit greater physiological activation and attentional concerns as compared to individuals with opposite characteristics (68). Fear of reinjury, anxiety or rumination about playing status, and inadequate performance compared to previous standards are common thoughts expressed by injured athletes (14, 59, 60). Appropriate interventions that involve techniques such as cognitive restructuring (e.g., reframing negative thinking patterns), and relaxation techniques (e.g., deep nasal breathing or visualization) can positively influence the stress response and mitigate the risk of future injury (3). Conversely, a resilient athlete (one who has overcome past injuries and has coping skills and social support) will likely have better adherence to rehabilitation programs and more positive outcomes (68, 69).

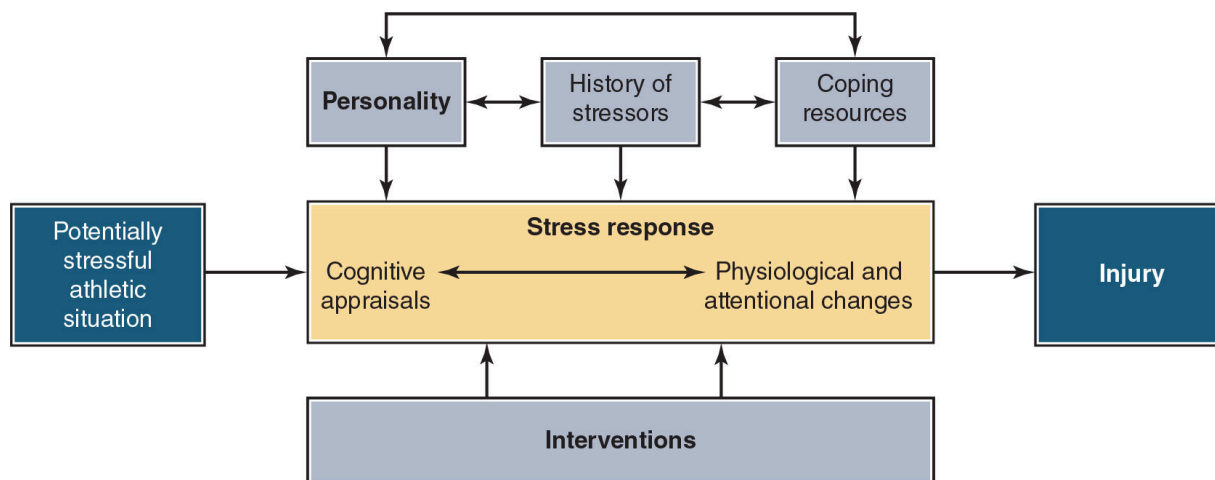


FIGURE 9.4 The stress and injury model.

Adapted by permission from J.M. Williams and M.B. Andersen, "Psychosocial Antecedents of Sport Injury: Review and Critique of the Stress and Injury Model," *Journal of Applied Sport Psychology* 10, no. 1 (1998): 5-25, reprinted by permission of the publisher (Taylor & Francis Ltd, <http://www.tandfonline.com>).

Psychological Factors That Can Affect an Athlete After Injury

Psychological concerns emerge throughout the process of injury recovery and often include timelines of rehabilitation to return to training and competition. Athletes must process a variety of conflicts once an injury has been sustained, including attending to and listening to the body, tending to the pain versus pushing through the pain, and interpersonal challenges and conflicts with teammates. Also, injuries can be a challenge to an athlete's identity and concerns about time left in the season or his or her career and falling behind teammates (60). Proper management of their relative effort and intensity to avoid future reinjury becomes a primary goal. Pushing through pain and injury may be falsely viewed as heroic and expected by teammates, parents, coaches, and fans (63).

The Cognitive Appraisal Model (66) proposes four key components that may influence how athletes judge their injury and how it affects their emotional and recovery response due to injury ([figure 9.5](#)). By understanding this model, strength and conditioning professionals can offer more support and guidance to help athletes return to their preexisting confidence and performance levels.

Personal Factors

The psychological impact of an injury on an athlete can be very specific to each individual. Strength and conditioning professionals can better understand how athletes will respond to injury by understanding these key personal factors:

- *History of injury.* Is this injury acute or chronic? Has the athlete experienced this injury before?
- *Athletic identity.* Does the athlete highly identify as an athlete and look to others for acknowledgment of that role? According to a 2022 study, almost 40% of student-athletes have a high athletic identity, strongly serving as a foundation of their self-worth (2).
- *Personality.* Is the athlete naturally confident and outgoing or introverted? Do they chronically worry about what they cannot accomplish or perform?
- *Pain tolerance.* How well does the athlete tolerate discomfort during training and rehabilitation?
- *Coping skills.* How has the athlete managed stressful situations in the past? Do they engage in positive (e.g., regular physical activity or relaxation strategies) or negative (e.g., risky activities or avoidant behaviors) coping strategies?
- *Age and sex.* Does the athlete have specific concerns due to their age (e.g., transportation) or sex (biological differences such as energy availability or hormonal differences)?

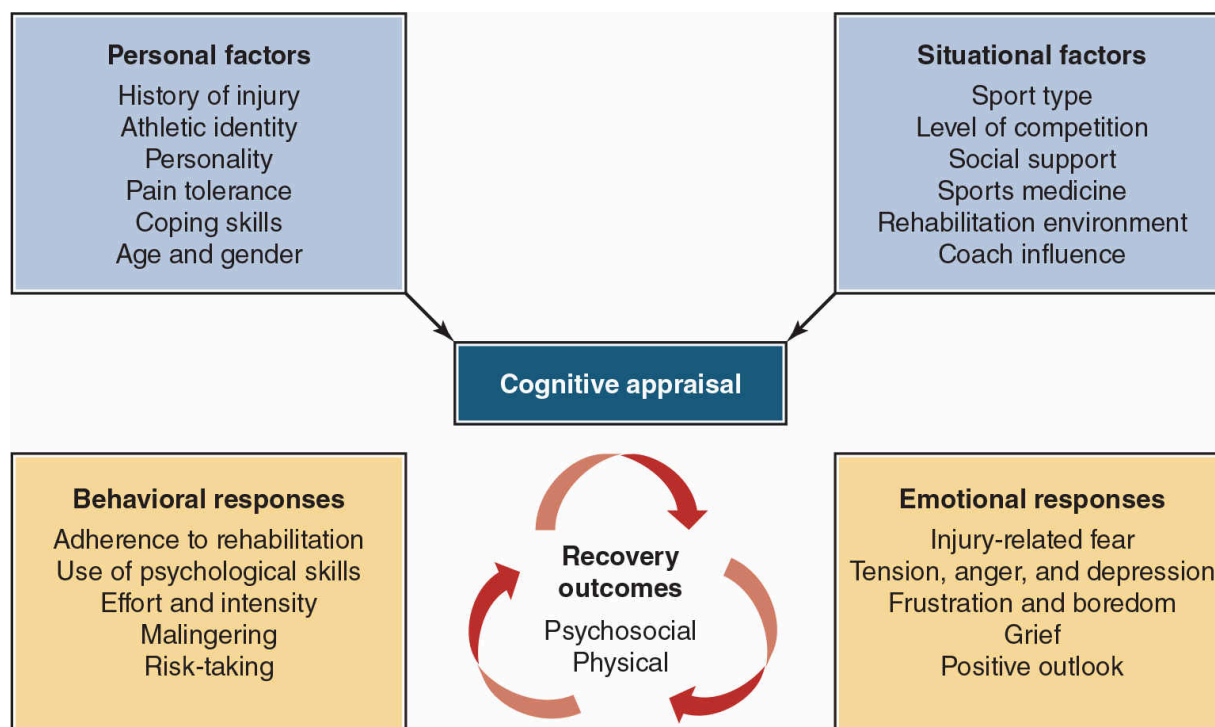


FIGURE 9.5 The cognitive appraisal model.

Reprinted by permission from D.M. Wiese-bjornstal, A.M. Smith, S.M. Shaffer, and M.A. Morrey, "An Integrated Model of Response to Sport Injury: Psychological and Sociological Dynamics," *Journal of Applied Sport Psychology* 10, no. 1 (1998): 46-69, reprinted by permission of the publisher (Taylor & Francis Ltd, <http://www.tandfonline.com>).

Situational Factors

In addition to personal factors, variability in environmental factors can influence how athletes view their injury and their return to sport, including the following:

- *Sport type*. Is it individual or team based? Is there a high risk of reinjury based on the sport?
- *Level of competition*. Is this a high school athlete or a professional athlete?
- *Social support*. Is the athlete surrounded by friends, family, and teammates for emotional and social support? Social support helps one learn that they are valued and supported regardless of athletic performance or the identity as a competitive athlete (63).
- *Sports medicine team*. Does the athlete have proper access to a professional rehabilitative team?

- *Rehabilitation environment.* Similarly, where is the athlete rehabilitating their injury? Does it require a lengthy commute? Is it adequate for proper sport-specific movements?
- *Coach influence.* Does the sport coach actively support the athlete during the rehab process? Is there pressure to return as soon as possible with the athlete fearing losing his or her position? Coaches should be attentive to the possibility that athletes could be struggling with challenges related to conflicts with coaches who may not understand the extent of their injuries; suggest that they are weak; and may explicitly minimize, invalidate, or dismiss the athlete's struggles (60).

Behavioral Responses

Consider how the athlete behaves when facing the impact of injury on his or her health and performance:

- *Adherence to rehabilitation.* Does the athlete demonstrate consistency and commitment to the active process?
- *Use of psychological skills.* Are techniques such as goal setting, imagery, and visualization integrated to improve their recovery timeline?
- *Effort and intensity.* How does the athlete's level of motivation change over time?
- *Malingering.* Does the athlete exacerbate pain or readiness levels to avoid future responsibilities or changes in their timeline?
- *Risk-taking.* Does the athlete engage in activities that have not yet been cleared for participation? Do they avoid certain activities because they feel they are not yet ready to perform?

Emotional Responses

Strength and conditioning professionals can become aware of the emotional responses that athletes may experience during the return-to-play process, including the following:

- *Injury-related fear.* How does the athlete embrace progressive overload into more sport-specific movements?
- *Tension, anger, depression.* Has the injury created conflict within the athlete's normal activities of daily living and sports participation?
- *Frustration and boredom.* Is the rehabilitation process mentally engaging and motivating?
- *Grief.* Was there a loss in a specific competition or season? Did the athlete lose his or her position or formal role with the team due to the injury?
- *Positive outlook.* Does the athlete view this setback as an opportunity to rebuild and improve?

From the lens of a strength and conditioning professional, athletes' responses to injury are subjective and multifaceted. It is likely that mental and physical readiness do not occur at the same rate. The psychological impact of an injury on an athlete influences more than weight room or athletic performance. For example, in a study of over 100 Division I American football players, academic stress (during regular examination periods) was associated with a significant increase in physical injuries (36). By understanding the pre- and post-injury psychological implications on mental and physical performance, strength and conditioning professionals can remain important members of the high-performance team and help develop a more robust and resilient athlete in the future.

What Strength and Conditioning Professionals Can Do to Help

As an integral member of the high-performance team, strength and conditioning professionals can continue to provide the necessary social, emotional, and physical support required for athletes (7, 45). Effective communication between sport coaches, strength and conditioning professionals, and sports medicine staff is imperative for

developing a culture that prioritizes athlete safety and maximizes performance (59).

Regular Communication With All Stakeholders

Strength and conditioning professionals should maintain regular communication with all athletic trainers, position coaches, parents (if appropriate), and head coaches. During the initial stage of injury rehabilitation, the strength and conditioning professional might contribute minimally to the rehab process. To ensure optimal transitions between phases of the rehabilitation program, the strength and conditioning professional should be aware of the training completed at the front end of the process. The sports medicine team can ensure their involvement is sequential and concurrent to prevent the risk of progressing too quickly and possibly reinjuring (59). Additionally, strength and conditioning professionals should maintain the injured athlete's privacy and confidentiality throughout the return-to-play process.

Specifically Tailored Training Programs

Depending on the severity of the injury, athletes could require a modified training program that may require different exercises, tempos, equipment, or arrangements (e.g., smaller, more personalized settings). If the time and resources allow it, every effort should be made to customize the training program to accommodate the athlete's current abilities and limitations.

Proper Goal Setting and Acknowledgment of Training Milestones

During a lengthy return-to-play timeline, setbacks, frustration, and lack of expedient progress are common. Coaches may consider overlapping appropriate goal-setting techniques with the sports medicine team to encourage and motivate the athlete. Goals should be relevant to controllable factors (e.g., Did I come to training? Did I perform ten repetitions?) and be flexible to prevent frustration from unexpected adjustments (63).

Progressive Integration Into Full-Group or Team Activities

One of the most basic psychological needs to enhance intrinsic motivation is relatedness, or a sense of belonging and connection (46). Progressively moving from privately scheduled, one-on-one rehabilitation sessions to larger team sessions may be an appropriate tactic to enhance the connection between the athlete and the team.

Improve Self-Efficacy and Motivation

Readiness to return to competition is a highly subjective and complex process. Characteristics of athletes who are ready to return to play demonstrate realistic expectations, high levels of self-efficacy, and low levels of anxiety (18). Coaches should minimize psychological barriers such as fear or anxiety by providing a psychologically safe and autonomy-supportive environment. Additionally, enhancing the athlete's self-efficacy about specific tasks (e.g., sprinting, jumping, cutting) and motivation to return to pre-injury levels would create a more intrinsically driven, process-oriented approach compared to an extrinsically driven, outcome-oriented approach to return to play (63).

Enhancing Motor Skill Acquisition and Learning

A comprehensive understanding of the impact and value of sport psychology on athletic performance is not complete without a working knowledge of the overlap of motor skill acquisition and learning topics. The integration of motor skill acquisition will improve both the athletes' performance and the ability of those who coach them. Selected techniques to enhance motor skill learning are discussed in the next sections, including those pertaining to practice schedule, instructions, and feedback. While research demonstrates favorable outcomes for particular techniques, it is important to consider the unique abilities of the athlete, the task, and the environment. What may facilitate learning for one athlete may have a different effect on another. The following sections can thus serve as a basic guide that can be tailored to the needs of the athlete, the task, and the environment.

Learning–Performance Distinction

Before considering techniques to enhance motor skill learning, it is important to recognize the learning–performance distinction. Learning is a process that results in a relatively permanent change in the capability to perform a motor skill (50). Performance is the execution of the skill in the current environment (29). Performance can be affected by arousal, motivation, and numerous other factors and therefore may not be indicative of the skill capabilities of the individual (9). While the techniques discussed here will facilitate skill learning of the individual, some might actually lead to performance decrements during the practice session. It is imperative to recognize this possibility and not assume that an athlete's performance during a given practice session reflects his or her learning of the given skill.

Practice Schedule

Practice is essential for motor skill learning. More importantly, challenging practice enables motor skill acquisition; mere repetitions alone are not enough to change behavior (43). One of the many ways to facilitate skill learning is through the manipulation of the practice structure and schedule.

Whole Versus Part Practice

With respect to teaching a complex motor skill, there is debate about the effectiveness of whole versus part practice. **Whole practice** addresses the skill in its entirety, whereas **part practice** separates the skill into a series of subcomponents. As a general rule, tasks that are challenging but have low interrelatedness of the subcomponents are learned better with part practice (40). For example, the snatch is a skilled power movement that can be broken down into four subcomponents: the first pull, transition, second pull, and catch. On the other hand, whole practice would tend to be favored for a task with subcomponents that are highly interrelated, because part practice can inhibit the effective regrouping of the subcomponents (40). In the instance of a lunge, it would be ineffective to separate the movements of the front leg from the movements of the lead leg, because these two components are highly interrelated. Therefore, a lunge would be better learned as a whole. Regardless of the task interrelatedness, in instances in which the skill may be dangerous or costly to learn as a whole, part practice is the favored choice.

Internal and External Cues for Coaching Feedback

The best coaching cues use the fewest words but make the greatest impact. Sometimes gestures or single words can be effective. It is important to stay brief with either motivation or instruction-related feedback during training sessions. If not, coaches tend to overcoach, a process known as *verbal overshadowing*, which can disrupt fluid motions by introducing cognitive processing. Athletes with limited experience may need time to experiment and learn to move and self-correct during

training. Often coaches are anxious to prevent injury, and overcoaching can disrupt the self-exploration process. By overcoaching, the athlete cannot self-correct, an important part of autonomous learning. Cues can be internal (“breathe,” “eyes up,” “back straight,” “heels on the floor”) and external (“push through the floor,” “try to pull down on the bar as you come up from the squat,” “make a letter C with your elbows as you rack the bar in the power clean”).

By focusing attention, coaches describe the mental intensity and effort needed for the specific technique. For example, lifting coach Gayle Hatch uses cues to teach stiff arms and trapezius contractions during the initial pull phase of the clean movement by saying, “When the elbows bend, the power ends” (G. Hatch, personal communication. March 15, 2007). Likewise, the cues for procedural movements and speed for the jerk movement can include “Dip and drive.” These represent technique and motivational cues. In the first instance, the instructions use external cues to enhance the lifting technique. In the second example, the coach uses motivational words to represent speed and intensity. Research has suggested that early in learning, external cues can be very effective. As the movements become refined and minor technique corrections are needed, internal cues may be more appropriate (76).

If part practice is to be used, a task can be separated into subcomponents in numerous ways. **Segmentation** breaks down the task into a series of subcomponents that have clear breaks between them (66). For example, as mentioned previously, the snatch can be broken down into the first pull, the transition, the second pull, and the catch. **Fractionalization** breaks the tasks into subcomponents that occur simultaneously (66). In performing the push press, the athlete practices the press motion of the arms and the push motion of the legs independently. **Simplification** adjusts the difficulty of the tasks by changing task characteristics such as the execution speed or the equipment used (66). In the snatch, the athlete first practices the subcomponents with a PVC pipe (or similar dowel rod). In this instance, both segmentation and fractionalization are used.

When teaching the subcomponents of the task, multiple methods can be used to integrate the parts back into the whole skill. **Pure-part training** (also known as the *part-whole* method) has the athlete practice each subcomponent of the skill multiple times independently. After all components have been practiced, the skill is practiced in its entirety (66). In a snatch, the first pull is practiced,

then the transition, followed by the second pull, and ending with the catch. After all skills have been practiced multiple times, the snatch is practiced. **Progressive-part training** has the athlete practice the first two parts in isolation before practicing these parts together (66). The athlete then practices the third subcomponent before practicing all three parts together. In the snatch, the athlete first practices the first pull and then the transition, then the first pull with the transition. The athlete then practices the second pull before practicing the first pull, transition, and second pull as a combined skill. This progression continues until the whole skill has been reintegrated. **Repetitive part training** has the athlete practice only the first part in isolation; then each subsequent part is added until the whole task is reintegrated (66). The athlete in this instance would first practice the first pull; then the first pull with the transition; then the first pull, transition, and second pull; and so forth. With use of simplification, characteristics of the task can be added gradually, increasing the difficulty of the task. The choice of sequencing will depend on the task and the goals of the given training session.

Random Practice

Traditionally, skills are practiced in a blocked fashion, such that the athlete practices the same skill multiple times before progressing to another skill. In **random practice**, multiple skills are practiced in a random order during a given practice session. For example, during blocked practice, the athlete performs multiple squat depth jumps before progressing to another skill. In random practice, an athlete may perform a squat depth jump, a depth jump with a lateral movement, a split squat jump, and a side-to-side push-off in a random order. The athlete continues to repeat these skills in a random order. While performance on each individual skill initially declines during random practice, learning is facilitated by this practice design (53). When applied in a sport setting, an integration of blocked and random practice (a few repetitive attempts at each

skill before proceeding) maximizes the benefits of both practice schedules (32). In the instance of the jumps, the athlete would perform a few repetitions of the squat depth jump before moving on to another movement such as the side-to-side push-off.

Variable Practice

Similar to random practice, **variable practice** includes variations of the same skill within a single practice session as opposed to specific practice in which a specific skill (i.e., depth jump to second box of a specific height) is repeated multiple times. In variable practice, the athlete would practice stepping from and jumping to boxes of varying heights. Similar to random practice, variable practice can impair performance during the practice session but can enhance performance on a novel variation of a skill (33) such as the ability to quickly jump to a second box after stepping off a box from an unpracticed height. A combination of specific and variable practice allows the athlete to develop sport-specific skills while also providing the athlete with the flexibility to perform in unfamiliar contexts (33). This flexibility is important to an athlete's success, because sport requires the athlete to perform in unfamiliar environments (travel games) and to accurately modify a practice skill in response to an opponent. As an example, training to land from and jump to varying heights will allow the athlete to respond in a game and execute the skill in an unfamiliar context.

Observational Learning

Observational practice (action observation), or practice through observation of the task or skill to be performed, has important implications for motor skill learning. Observational practice frequently uses prerecorded videos or live demonstrations. The individual executing the observed skill can be a novice or a competent or expert performer. When physical practice is combined with observational practice, learning is enhanced (52). In a strength and conditioning setting, partner work can facilitate learning. As one

athlete completes a lift or a drill, the other athlete can observe during his or her rest period and vice versa. Sakadjian and colleagues (49) found that observational practice combined with physical practice facilitated improved power clean technique as compared to physical practice alone. Observational practice can assist athletes in achieving the desired technique, thus maintaining athlete safety and allowing for a more rapid progression to heavier lifting.

Instructions

A coach can facilitate learning by varying the amount of detail provided in the instruction to the athletes' strongest learning style. Instructional styles can be divided into explicit instructions, guided discovery, and discovery. **Explicit instructions** include prescriptive information that gives the athlete the "rules" for effectively executing the given task. In learning a squat, the athlete receives detailed instructions about the specific body position throughout the movement as well as the type (e.g., flexion–extension) and amount of movement at each of the joints. **Guided discovery** provides the athlete with instructions about the overall movement goal and important prompts for task accomplishment without explicitly telling the athlete how to accomplish the task. The athlete is told the goal of the squat depth jump and a few key reminders to prevent injury, such as maintaining a neutral spine. This style provides some direction while allowing the athlete to explore relationships between the executed movement patterns and the associated movement goal. Finally, **discovery** instructs the athlete on the overarching goal of the task and the athlete receives little to no direction. In this instance the athlete is just told to squat to a particular depth and is allowed to explore methods to accomplish this task. A discovery instructional style can slow the learning process, while explicit instructions can impair performance in a stressful environment (56). The decreased attentional demands of discovery and guided

discovery instructional styles allow athletes to increase their focus on the task-relevant cues related to task execution (75).

Feedback

Feedback plays a significant role in motor skill acquisition by providing the athlete with information about the movement pattern and the associated goal. This information can then be used to make appropriate adjustments to achieve the desired movement pattern and goal. **Intrinsic feedback** is feedback provided to the athlete by the athlete from the senses—for example, sensory information about missing a box during a squat box jump. Integration of the sensory information allows the athlete to fine-tune and adjust the movement pattern to produce the desired movement and associated task goal. **Augmented feedback** is feedback provided to the athlete by either an observer, such as a coach, or technology, such as video or laboratory equipment. After the squat box jump, a coach can tell the athlete that the hips' countermovement was too slow, resulting in missing the box.

The remainder of this section focuses on augmented feedback. Augmented feedback can be broken down into knowledge of results and knowledge of performance. **Knowledge of results** provides the athlete with information about the execution of the task goal. For example, with a T-drill, the coach can tell the athlete how quickly he or she completed the drill. The athlete can also be given normative information about how this time compares with others. Both positive normative feedback and feedback without comparison augment skill learning compared to negative normative feedback (34). **Knowledge of performance** feedback provides the athlete with information about his or her movement pattern. This can be delivered in the form of video analysis or through the use of specialized laboratory equipment such as a force plate. In the instance of the T-drill, the coach gives the athlete specific information about the movement during the T-test such as body position during changes of direction. When the task goal is a

movement outcome—for example, proper form on a power clean—there is an overlap between knowledge of results and knowledge of performance and thus they can become one and the same.

The timing and frequency of feedback also influence learning outcomes. Feedback that is offered concurrently with the task enhances performance while impairing learning; therefore, this feedback is beneficial in a competition setting (69). However, feedback that is provided after task execution facilitates skill learning (69). This feedback can be provided either after every trial or after a series of trials. While decreased frequency impairs performance during practice, less frequent feedback enhances skill acquisition as compared to feedback after every trial (70), except in instances in which the skill is complex (74). Therefore, in initial skill learning it would be beneficial to provide more frequent feedback that decreases as athletes progress in their skill level.

Feedback can be used to facilitate both learning and performance. The timing and frequency of the feedback have different influences on performance and learning. While concurrent feedback is beneficial for competition, delayed feedback that is initially frequent and decreases with time will facilitate learning of complex movement patterns.

Conclusion

Although a scientific and motivationally sound coaching program can greatly help in the development of athletes, several other complementary psychological techniques can enhance their overall effectiveness. In reality, most athletes probably have had both adaptive and maladaptive experiences in the form of countless practices and competitions and, as a result, have developed a more or less effective mental skills repertoire. A working understanding of some of the psychological principles and tools for the generation of cognitive-behavioral change included in this chapter may help facilitate improved performance and an enhanced quality of life.

The coach and athlete can each contribute to improved performance outcomes and increase the enjoyment of competition by attending to the psychological aspects of instruction and performance. A positive, goal-oriented coaching approach is one of the most powerful contributors to psychological preparation for sport. The physical and nutritional preparation of the athlete represents the foundation on which performance potential is based, and the role of psychology is to mentally manage the developed physical resources (i.e., strength, speed, flexibility, and skill), thereby allowing the athlete to achieve his or her potential on a more consistent basis. Additionally, an adequate understanding of the mind-body relationship, as reflected in the constructs of this chapter, can facilitate communication between the strength and conditioning professional and the athlete and aid the athlete in controlling and managing emotion, arousal, focus, and motivation. Using appropriate psychological techniques can help this self-management process. The experience of success in sport may be important in and of itself, but the greater outcome of improved mental skills use to enhance self-esteem, confidence, and positive self-concept improves the quality of life in general.

KEY TERMS

achievement motivation
anxiety
arousal
athletic identity
attention
augmented feedback
autogenic training
catastrophe theory
cognitive anxiety
counterconditioning
diaphragmatic breathing
discovery
distress
drive theory
emotions
eustress
explicit instructions
extrinsic motivation
fight-or-flight
flow
focal dystonia
fractionalization
goal setting
guided discovery
ideal performance state
imagery
individual zones of optimal functioning
intrinsic feedback
intrinsic motivation
inverted-U theory
knowledge of performance
knowledge of results
long-term goals
mental health

mental imagery
motivation
motive to achieve success (MAS)
motive to avoid failure (MAF)
negative punishment
negative reinforcement
observational practice
operant
outcome goals
part practice
physiological efficiency
positive punishment
positive reinforcement
process goals
progressive muscle relaxation (PMR)
progressive-part training
psychological efficiency
pure-part training
random practice
readiness
repetitive part training
reversal theory
routine
segmentation
selective attention
self-confidence
self-controlled practice
self-efficacy
self-talk
short-term goals
simplification
somatic anxiety
sport psychology
state anxiety

stress
stressor
systematic desensitization (SD)
trait anxiety
variable practice
whole practice

STUDY QUESTIONS

1. Which of the following is the best definition of mental health?
 - a. the ability to perform well under pressure and avoid emotional expression
 - b. a state of emotional, psychological, and social well-being
 - c. the absence of any diagnosed mental illness
 - d. a personality trait that dictates an athlete's motivation status
2. What is the first step in identifying psychological signs of distress in an athlete?
 - a. referring the athlete to a mental health professional
 - b. asking the athlete to complete a standardized mood questionnaire
 - c. monitoring the athlete's training load and recovery scores
 - d. classifying the athlete's affect, behavior, and cognitions
3. Which of the following is the distinguishing factor of depression from anxiety?
 - a. hopelessness or helplessness
 - b. rumination
 - c. worry
 - d. circular thinking
4. Predisposing psychological factors to injury include all the following except which option?
 - a. history of stressors
 - b. violent outbursts
 - c. personality

- d. coping resources
5. Which of the following best describes a key personal factor that can influence how an athlete psychologically responds to an injury?
- a. type of sport that the athlete competes in
 - b. availability of a professional rehabilitative team
 - c. athlete's history of previous injuries and coping mechanisms
 - d. athlete's competitive experience